

Fock Space:

$$\mathcal{H}_0 = \text{Span}\{|vac\rangle\}$$

$$\mathcal{H}_1 = \text{Span}\{|n\rangle\}_n$$

$|n\rangle$

$|n\rangle, |m\rangle$

$|m\rangle$

$$\mathcal{H}_2 = \text{Span}\{|n\rangle \otimes |m\rangle\}_{n,m}$$

$$|F := \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \dots$$

$$\frac{\underbrace{|1\rangle}_{\in \mathcal{H}_1} + \underbrace{|2\rangle \otimes |3\rangle}_{\in \mathcal{H}_2}}{\sqrt{2}} \in |F$$

occupation number representation :
ONR

$$|10203\dots\rangle : \begin{array}{lcl} 1 & \text{particle in} & |1\rangle \\ 0 & \text{"} & |2\rangle \\ 2 & \text{"} & |3\rangle \\ & & \vdots \end{array}$$

ex: $\left\{ \begin{array}{l} |vac\rangle \\ |1\rangle \\ |2\rangle \\ \frac{|1\rangle \otimes |2\rangle - |2\rangle \otimes |1\rangle}{\sqrt{2}} \end{array} \right.$

ONR $\rightarrow$

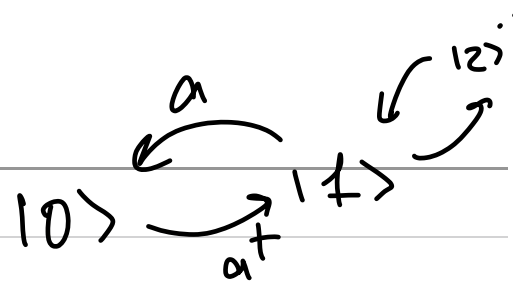
$$\begin{array}{lcl} |vac\rangle & & \\ |1\rangle & \rightarrow & |140\rangle \\ |2\rangle & \rightarrow & |041\rangle \\ \frac{|1\rangle \otimes |2\rangle - |2\rangle \otimes |1\rangle}{\sqrt{2}} & \rightarrow & |111\rangle \end{array}$$

0'th level $\rightarrow$ Quantum system $\rightarrow \text{Span}\{|1\rangle, |2\rangle, |3\rangle, \dots\}$

$$|1\rangle \otimes |3\rangle \otimes |0\rangle \otimes \dots = |130\dots\rangle$$



Creation & annihilation operators:



$$\langle \text{vac} | \text{vac} \rangle = 1$$

$$\underline{a_\mu^+ |\text{vac}\rangle = |\mu\rangle} \rightarrow \text{single-particle state } \mu.$$

$$a_\mu^- := (a_\mu^+)^{\dagger} \rightarrow \langle \mu | \mu \rangle = \langle \text{vac} | \boxed{a_\mu^- a_\mu^+} |\text{vac}\rangle = 1$$

$$\begin{aligned} \hookrightarrow a_\mu^- a_\mu^+ |\text{vac}\rangle &= |\text{vac}\rangle \\ \underbrace{a_\mu^- |\mu\rangle} &= |\text{vac}\rangle \end{aligned}$$

$$\underline{a_\mu^- |\text{vac}\rangle = 0}$$

Fermions:

$$(a_\mu^+)^2 = 0$$

$$(a_\nu^+)^2 = 0$$

$$\frac{|\mu\rangle + |\nu\rangle}{\sqrt{2}} = \frac{a_\mu^+ + a_\nu^+}{\sqrt{2}} |\text{vac}\rangle$$

$$\underline{(a_\mu^+ + a_\nu^+)^2 = 0} \rightarrow \cancel{(a_\mu^+)^2} + \cancel{(a_\nu^+)^2} + a_\mu^+ a_\nu^+ + a_\nu^+ a_\mu^+ = 0$$

$$a_\mu^+ a_\nu^+ + a_\nu^+ a_\mu^+ = 0 \rightarrow \underline{\{a_\mu^+, a_\nu^+\} = 0}$$

$$a_\mu^+ a_\nu^+ |\text{vac}\rangle = -a_\nu^+ a_\mu^+ |\text{vac}\rangle$$

$$N_\mu := a_\mu^+ a_\mu^- \rightarrow N_\mu |\text{vac}\rangle = 0 \rightarrow \text{E value } 0$$

$$a_\mu^- a_\mu^+ |\text{vac}\rangle = |\text{vac}\rangle \rightarrow \underbrace{a_\mu^+ a_\mu^- a_\mu^+}_{N_\mu} |\text{vac}\rangle = a_\mu^+ |\text{vac}\rangle$$

$$N_\mu (a_\mu^+ |vac\rangle) = 1 (a_\mu^+ |vac\rangle)$$

$\underbrace{\hspace{10em}}_{E \text{ value } 1}$

$$N'_\mu := a_\mu^- a_\mu^+ \rightarrow N'_\mu |vac\rangle = |vac\rangle$$

$$N'_\mu (a_\mu^+ |vac\rangle) = 0$$

$\underbrace{a_\mu^- a_\mu^+}_{0}$

$$\left. \begin{array}{l} N'_\mu |vac\rangle = |vac\rangle \\ N'_\mu (a_\mu^+ |vac\rangle) = 0 \end{array} \right\} N'_\mu = \mathbb{1} - N_\mu$$

$$a_\mu^+ a_\mu^- + a_\mu^- a_\mu^+ = \mathbb{1} \Rightarrow \{a_\mu^+, a_\mu^-\} = \mathbb{1}$$

in $|n\rangle$ representation: $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$a_\mu^+ = \overset{\uparrow}{\sigma_z} \otimes \sigma_z \otimes \dots \otimes \sigma_+ \otimes \mathbb{1} \otimes \dots \otimes \mathbb{1}$$

$$a_2^+ |1 \ 0 \ 1\rangle = a_2^+ (a_1^+ a_3^+ |vac\rangle) = -a_1^+ a_2^+ a_3^+ |vac\rangle$$

$$= -|1 \ 1 \ 1\rangle$$

$$a_3^+ |1 \ 1 \ 0 \ 1\rangle = a_3^+ (a_1^+ a_2^+ a_4^+ |vac\rangle) = |1 \ 1 \ 1 \ 1\rangle$$

Bosons:

$$\hat{N}_\mu | \dots n_\mu \dots \rangle := n_\mu | \dots n_\mu \dots \rangle$$

$$N_\mu a_\nu^\pm | \dots n_\mu \dots \rangle = (n_\mu \pm \delta_{\mu\nu}) a_\nu^\pm | \dots n_\mu \dots \rangle$$

$$[a_\mu^+, a_\nu^+] = 0 = [a_\mu^-, a_\nu^-]$$

$$[N_\mu, a_\nu^\pm] |\dots n_\mu \dots n_\nu \dots\rangle$$

$$\begin{aligned} \underbrace{N_\mu a_\nu^\pm}_{\text{blue underline}} |\dots n_\mu \dots\rangle &= (n_\mu \pm \delta_{\mu\nu}) a_\nu^\pm |\dots n_\mu \dots\rangle \\ &= a_\nu^\pm \underbrace{n_\mu |\dots n_\mu \dots\rangle}_{N_\mu |\dots n_\mu \dots\rangle} \pm \delta_{\mu\nu} a_\nu^\pm |\dots n_\mu \dots\rangle \end{aligned}$$

$$[N_\mu, a_\nu^\pm] |\dots n_\mu \dots\rangle = \pm \delta_{\mu\nu} a_\nu^\pm |\dots n_\mu \dots\rangle$$

$$[N_\mu, a_\nu^\pm] = \pm \delta_{\mu\nu} a_\nu^\pm$$

$$\begin{aligned} [N_\mu, a_\nu^+ a_\nu^-] &= [N_\mu, a_\nu^+] a_\nu^- + a_\nu^+ [N_\mu, a_\nu^-] \\ &= \delta_{\mu\nu} a_\nu^+ a_\nu^- - \delta_{\mu\nu} a_\nu^+ a_\nu^- \\ &= \delta_{\mu\nu} [a_\nu^+, a_\nu^-] \end{aligned}$$

$$\mu \neq \nu: [N_\mu, a_\nu^+ a_\nu^-] = 0$$

$$N = \begin{pmatrix} * & & & \\ & * & & \\ & & * & \\ & & & * \dots \end{pmatrix} \quad a^+ = \begin{pmatrix} 0 & * & & \\ & 0 & * & \\ & & 0 & * \\ & & & 0 \dots \end{pmatrix}$$

$$a_\mu^+ = \sum_{\dots, n_\mu} \lambda_{n_\mu+1}^* |\dots n_\mu+1 \dots\rangle \underbrace{\langle \dots n_\mu \dots |}_{\text{red underline}}$$

$$a_\mu^- = \sum_{\dots, n_\mu} \lambda_{n_\mu+1}^* \underbrace{|\dots n_\mu \dots\rangle}_{\text{red underline}} \langle \dots n_\mu+1 \dots |$$

$$\begin{aligned}
 a_{\mu}^+ a_{\mu}^- &= \sum_{\substack{\dots, n_{\mu}=0 \\ \infty}}^{\infty} |\lambda_{n_{\mu}+1}|^2 |\dots n_{\mu}+1 \dots\rangle \langle \dots n_{\mu}+1 \dots| \\
 &= \sum_{\substack{\dots=0 \\ n_{\mu} \geq 1}}^{\infty} |\lambda_{n_{\mu}}|^2 |\dots n_{\mu} \dots\rangle \langle \dots n_{\mu} \dots|
 \end{aligned}$$

$$\lambda_n := \sqrt{n} \rightarrow \underline{\underline{a_{\mu}^+ a_{\mu}^- = N_{\mu}}}$$

$$a_{\mu}^+ |\dots n_{\mu} \dots\rangle = \sqrt{n_{\mu}+1} |\dots n_{\mu}+1 \dots\rangle$$

$$a_{\mu}^- |\dots n_{\mu} \dots\rangle = \sqrt{n_{\mu}} |\dots n_{\mu}-1 \dots\rangle$$

$$|\dots n_{\mu} \dots\rangle = \prod_{\mu} \frac{(a_{\mu}^+)^{n_{\mu}}}{\sqrt{n_{\mu}!}} |Vac\rangle$$

$$|1\rangle + |2\rangle - |3\rangle = a^+ |0\rangle + \frac{a^+ a^+}{\sqrt{2}} |0\rangle - \frac{a^+ a^+ a^+}{\sqrt{6}} |0\rangle$$

$$\left(a^+ + \frac{(a^+)^2}{\sqrt{2}} - \frac{(a^+)^3}{\sqrt{6}} \right) |Vac\rangle$$